LUCAS RICHARD FATH HENNEMAN, PH.D.

Associate Professor of Environmental Engineering
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Sid and Reva Dewberry Department of Civil, Environmental and Infrastructure Engineering
College of Engineering and Computing • George Mason University • Fairfax, VA 22030

Academic Degrees

Ph.D., Environmental Engineering, Georgia Institute of Technology 2017
Thesis: *Air Pollution Accountability: Assessing Regulatory Impacts on Emissions and Air Quality*
Advisor: Dr. Armistead G. Russell

M.S., Environmental Engineering, Georgia Institute of Technology 2014

B.S., Environmental Engineering, Johns Hopkins University with general honors 2012

Academic Positions

Associate Professor of Environmental Engineering 2026 – present
Assistant Professor of Environmental Engineering 2020 – 2026
Sid & Reva Dewberry Department of Civil, Environmental, and Instructure Engineering
Volgenau School of Engineering
College of Engineering and Computing
George Mason University

Visiting Scientist 2020 – 2023
Department of Biostatistics
T.H. Chan School of Public Health
Harvard University

Postdoctoral Fellow 2017 – 2020
Departments of Biostatistics and Environmental Health
T.H. Chan School of Public Health
Harvard University

Academic and Professional Honors

- Selected as GeoCAFE Scholars Program 2025 cohort member, funded by NSF grant 23-529 and the CAFE Research Coordinating Center (NIH grant U24ES035309)
- Our article, "Inequitable Exposures to U.S. Coal Power Plant–Related $PM_{2.5}$: 22 Years and Counting" listed in *Environmental Health Perspectives* Editors' Choice Collection 2023. https://ehp.niehs.nih.gov/curated-collections/eds-choice-2023
- The Health Effects Institute Rosinbleth Early Career Research Awardee (2021)
- Georgia Tech School of Civil and Environmental Engineering Best PhD Thesis Award (2017)
- National Science Foundation Graduate Research Fellowship (2012-2017)
- Georgia Tech Association of Environmental Engineers and Scientists Outstanding PhD Student Award (2017)
- Georgia Tech President's Fellowship (2012-2016)

- Air and Waste Management Association Scholarship (2016)
- International Institute for Applied Systems Analysis (IIASA) Southern African Young Scientists Summer Program (2014-2015)
- Air and Waste Management Georgia Chapter Scholarship (2014)
- Air and Waste Management Association Southern Section Scholarship (2014)
- Johns Hopkins University Department of Geography and Environmental Engineering Faculty Award for Service and Academic Achievement (2012)
- Trans-Atlantic Program for Young Technology Leaders Delegate to Germany for the German-American Chamber of Commerce (2011)
- Thomas J. Watson Memorial Scholarship (2008-2012)

Refereed Journal Papers

Total citations: 1,429; *h-index*: 23 [Google Scholar]
Underline denotes student or postdoc in my group. IF stands for *impact factor*.

Papers published while at Mason

1. Rasel, Munshi Md, Cohan, D., Tong, D., and **Henneman, L.R.F.** (2026) "Coal power plants continue to offer route to meet new us $PM_{2.5}$ standards." *Environmental Science and Technology.* Link [IF = 11]
2. Ghosh, Chaudhuri, Mukherjee, Samanta, Saha, **Henneman, L.R.F.,** Majumdar, Sengupta, Chakraborty, Maiti, Ghosh, Biswas, Sinha. (2025) "Seasonal fluctuations in ambient particulate matter2.5 exposure differentially regulate JAK2/STAT3 signaling in never smoking rural and urban cohorts." *Free Radical Biology and Medicine.* Link. [IF = 8.2]
3. Ghosh, Chaudhuri, Mukherjee, **Henneman, L.R.F.,** Samanta, Saha, Sengupta, Sinha. (2025) "Ghosh, Chaudhuri, Mukherjee, Samanta, Saha, Henneman, L.R.F., Majumdar, Sengupta, Chakraborty, Maiti, Ghosh, Biswas, Sinha. (2025) "Seasonal fluctuations in ambient particulate matter2.5 exposure differentially regulate JAK2/STAT3 signaling in never smoking rural and urban cohorts." *Scientific Reports*. Link. [IF = 3.9]
4. Guha, S. and **Henneman, L.R.F.** (2025). "Intercomparison of data products for studying trends in $PM_{2.5}$ and ozone air quality over space and time in China: Implications for sustainable air quality management." *Sustainability*. Link. [IF = 3.3]
5. Shan, X., Casey, J.A., Shearston, J.A., **Henneman. L.R.F**. (2025). "Source-Specific Air Pollution in Areas and Periods with Limited Data: application to 1940 in the United States." *Environ. Res.: Health* Link.
6. **Henneman, L.R.F.**, Ryah Nadjafi, Xiaorong Shan, Jenna Krall. (2025). "Source-specific air pollution emissions inequalities from 2011-2020 in Virginia." *GeoHealth*. Link. [IF = 4.3]
7. Guidi, G., Dominici, F., Steinsultz, N., Dance, G., **Henneman, L.R.F.**, Richardson, H., Castro, E., Bargagli-Stoffi, F., Delaney, S. (2025). "The environmental burden of the United States' bitcoin mining boom." *Nature Communications*. Link [IF = 17]
8. Shan, X., Casey, J.A., Shearston, J.A., **Henneman. L.R.F.** (2024). "Methods for Quantifying Source-Specific Air Pollution Exposure to Serve Epidemiology, Risk Assessment, and Environmental Justice." *GeoHealth*. Link [IF = 4.3]
9. Guha, S., Zhang, T., Kinney, P., **Henneman, L.R.F.** (2024). "Fine particulate matter and ozone variability with regional and local meteorology in Beijing, China." *Atmospheric Environment. Link* [IF = 6]
10. Krall, J.R., Thornburg, J., Zhang, T., Pollack, A., Lee, Y., McCombs, M., **Henneman, L.R.F.** 2024. "Short-term associations between road features and in-vehicle $PM_{2.5}$ during daily trips." *Environments*. Link [IF = 3]

11. Zhang, T., Yan, B., **Henneman, L.R.F.**, Kinney, P., Hopke, K. (2024). "A review of recent source apportionment studies in China." *Science of the Total Environment.* Link [IF = 9.8]
12. Zheng, H., Li, S., Jiang, Y., Dong, Z., Yin, D., Zhao, B., Wu, Q., Liu, K., Zhang, S., Wu, Z., Wen, Y., Xing, J., **Henneman, L.R.F.**, Kinney, P.L., Hao, J., Wang, S., (2024). "Unpacking the Factors Contributing to Changes in $PM_{2.5}$-Associated Mortality in China from 2013 to 2019." *Environment International*. Link [IF = 13]
13. **Henneman, L.R.F.**, Choirat, C., Dedoussi, I., Dominici, F., Roberts, J., Zigler, C. (2023). "Mortality risk from United States coal electricity generation." *Science*. Vol 382, Issue 6673. pp. 941-946. Link [IF = 57]
14. Rasel, Munshi Md, Chen, K., Nethery, R., **Henneman. L.R.F.** (2023). "COVID-19 and Beyond: COVID-19 Interventions and Power Plant Emissions in the United States." *ES&T Engineering*. Link [IF = 7]
15. Nunez, Y., Benavides, J., Shearston, J., Daouda, M., **Henneman, L.R.F.**, Martin, R.V., McDuffie, E.E., Goldsmith, J., Casey, J., Kioumourtzoglou, M.A. (2023). "Air Pollution Emissions Trends in the United States from 1970 to 2010: An Environmental Justice Analysis." *Nature Communications*. Link [IF = 17]
16. Ebelt, S., Baxter, L., Erickson, H.S., **Henneman, L.R.F.**, et al. (2023). "Air Pollution Accountability Research: Moving from a Causal Chain to a Web." *Global Epidemiology*. Link [IF = 4]
17. Ortiz, L.E., Stiles, R., Whitaker, S., Maibach, E., Kinter, J., **Henneman, L.R.F.**, Krall,, J. Bubbosh, P., Cash, B. (2023). "Public health benefits of zero-emission electric power generation in Virginia." *Helion*. Link [IF = 4]
18. Chen, C., Ilango, S.D., **Henneman, L.R.F.**, Casey, J.A. Benmarhnia, T. (2023). "The local impacts of coal and oil power plant retirements on air pollution and cardiorespiratory health in California: An application of generalized synthetic control method." *Environmental Research.* Link [IF = 8]
19. **Henneman, L.R.F.**, Rasel, Munshi Md, Choirat, C., Anenberg, S., Zigler, C.M. (2023), "Inequitable Exposures to U.S. Coal Power Plant–Related $PM_{2.5}$: 22 Years and Counting." *Environmental Health Perspectives. DOI 10.1289/EHP11605.* [IF = 10]
20. Daouda, M., **Henneman, L.R.F.**, Goldsmith, J., Kioumourtzoglou, M., Casey, J. (2022). Racial/Ethnic Disparities in Nationwide $PM_{2.5}$ Concentrations: Perils of Assuming a Linear Relationship. *Environmental Health Perspectives*. Link [IF = 10]
21. Wikle, N., Hanks, E., **Henneman., L.R.F.**, Zigler, C.M. (2022) "A mechanistic model of annual sulfate concentrations in the United States." *Journal of the American Statistical Association.* Link [IF = 4]
22. Dey, T., Tyagi, P., Sabath, B., Kamareddine, L., **Henneman., L.R.F.**, Braun, D., Dominici, F. (2021) "Counterfactual time series analysis of short-term change in air pollution following the COVID-19 state of emergency in the United States." *Scientific Reports*. Link [IF = 5]
23. Chen, K., **Henneman, L.R.F.**, Nethery, R. (2021) "Differential impacts of COVID-19 Lockdowns on $PM_{2.5}$ across the United States." *Environmental Advances*. Link [IF = 5]
24. Daouda, M., **Henneman, L.R.F.**, Kioumourtzoglou, M., Gemmill, A., Zigler, C.M., Casey, J. (2021) "Association between county-level coal-fired power plant pollution and racial disparities in preterm births from 2000 to 2018." *Environmental Research Letters.* Link [IF = 7]
25. **Henneman, L.R.F.**, Shen, H., Hogrefe, C., Russell, A., Zigler, C.M. (2021) "Four decades of United States mobile source pollutants: spatial-temporal trends assessed by ground-based monitors, air quality models, and satellites." *Environmental Science and Technology.* Link [IF = 11]

Manuscripts in review

1. Bohloul, M., Rasel, M.M., Hildebrandt-Ruiz, L., Allen, D., Kimura, Y., Graves, J.D., **Henneman, L.R.F.** "Spatiotemporal Exposure Assessment of Air Pollution from Unconventional Oil and Gas Development in the Eagle Ford Shale, Texas, Using Multiple Dispersion Models"
2. Shearston, J., Shan, X., … **Henneman, L.R.F.**, et al. "Still separate and unequal: segregation and air pollution burdens in 1940 and 2010 in the United States."

3. Zhang, T., Hussein, S., Hopke, P., Zigler, C., Tong, D., Anenberg, S., **Henneman, L.R.F.H.** "Nationwide source apportionment reveals increasing importance of carbonaceous PM2.5 and sources with high spatial variability"

Papers published before Mason employment

1. **Henneman, L.R.F.**, Dedoussi, I.C., Casey, J.A., Barett, R.H., Choirat, C., and Zigler, C.M. (2020) "Comparisons of simple and complex methods for quantifying exposure to point source air pollution emissions." *Journal of Exposure Science and Environmental Epidemiology.* Link [IF = 5]
2. Casey, J.A., Su, J.G., Barrett, M.A., **Henneman, L.R.F.**, Zigler, C.M., *et al.* (2020). "Improved asthma outcomes observed in the vicinity of coal power plant retirement, retrofit and conversion to natural gas." *Nature Energy.* Link [IF = 57]
3. Kim, C., **Henneman, L.R.F.**, Choirat, C., and Zigler, C.M. (2020). "Health Effects of Power Plant Emissions through Ambient Air Quality." *Journal of the Royal Statistical Society*. Link [IF = 5]
4. Liao, S., **Henneman, L.R.F.**, and Zigler, C.M. (2020) "Posterior predictive treatment assignment methods for causal inference in the context of time-varying treatments." *Epidemiological Methods.* Link [IF = 1]
5. **Henneman, L.R.F.**, Mickley, L.J., and Zigler, C.M. (2019) "Air pollution accountability of energy transitions: the relative importance of wind fields and emissions in exposure changes." *Environmental Research Letters.* Link [IF = 7]
6. **Henneman, L.R.F.**, Choirat, C., and Zigler, C.M. (2019). "Accountability Assessment of Health Improvements in the United States Associated with Reduced Coal Emissions Between 2005 and 2012." *Epidemiology.* Link [IF = 5]
7. Shen, H., Chen, Y., Li, Y., Russell, A., Hu, Y., **Henneman, L.R.F.**, *et al.* (2019). "Relaxing energy policies on top of climate change will significantly undermine states' efforts to attain U.S. ozone standards." *One Earth.* Link [IF = 17]
8. Yu, X., Stuart, A., Liu, Y., Ivey, C., Russell, A., Kan, H., **Henneman, L.R.F.**, Sarnat, S., Hasan, S., Sadmani, A., Yang, X., Yu, H. (2019), "On the potential of Google Maps location history data to characterize historical individual mobility for retrospective air pollution health studies." *Environmental Pollution.* Link [IF = 9]
9. **Henneman, L.R.F.**, Liu, C., Chang, H., Mulholland, J., Tolbert, P., Russell, A. (2019) "Emissions reductions and air quality responses in Atlanta, Georgia in response to regulations: developing daily time series of changes in ten pollutants and uncertainties." *Environment International*. Link [IF = 13]
10. Qian, Y., **Henneman, L.R.F.**, Mulholland, J.A., Russell, A.G. (2019). "Empirical Development of Ozone Isopleths: Applications to Los Angeles." *Environmental Science and Technology Letters.* Link [IF = 11]
11. Abrams, J., Klein, M., **Henneman, L.R.F.**, Sarnat, S., Chang, H., Strickland, M., Mulholland, J.A., Russell, A.G., Tolbert, P. (2019). "Impact of air pollution control policies on cardiorespiratory emergency department visits, Atlanta, GA, 1999-2013." *Environment International.* Link [IF = 13]
12. **Henneman, L.R.F.**, Ivey, C., Choirat, C., Cummiskey, K., and Zigler, C.M. (2019). "Characterizing population exposure to coal emissions sources in the United States using the HyADS model." *Atmospheric Environment*. Link [IF = 6]
13. Lawal, A.S., Guan, X., Liu, C., **Henneman, L.R.F.**, Bhogineni, V., Weber, R.J., Nenes, A., and Russell, A. (2018). "Linked Response of aerosol acidity and Ammonia to SO_2 and NO_X Emissions Reductions in the US." *Environmental Science and Technology*. Link [IF = 11]
14. Shen, H., Chen, Y., Chen, Y., Fan, R., Li, Y., Zeng, E., Huang, Y., Yu, H., Shen, G., Shen, H., Zhong, Q., Russell, A., **Henneman, L.R.F.**, Ru, M., Zou, Y., Hu, Y. (2018). "Impacts of rural worker migration on air quality and health in China: From the perspective of upgrading residential energy consumption." *Environment International*. Link [IF = 13]
15. **Henneman, L.R.F.**, Liu, C., Shen, H., Hu, Y., Mulholland, J., Russell, A. (2017). "Responses in ozone and its production efficiency attributable to recent and future emissions changes in the eastern United States." *Environmental Science and Technology*. Link [IF = 11]

16. **Henneman, L.R.F.**, Liu, C., Hu, Y., Mulholland, J., Russell, A. (2017). "Air quality modeling for accountability research: Operational, dynamic, and diagnostic evaluation." *Atmospheric Environment*. Link [IF = 6]
17. **Henneman, L.R.F.**, Chang, H., Lavoué, D., Mulholland, J., Russell, A. (2017). "Assessing regulatory impacts on ozone and $PM_{2.5}$ using empirically- and air quality model-derived pollutant sensitivities." *Air Quality, Atmosphere, and Health.* Link [IF = 5]
18. **Henneman, L.R.F.**, Liu, C., Mulholland, J., Russell, A. (2016). "Evaluating the effectiveness of air quality regulations: A review of accountability studies and frameworks." *Journal of the Air and Waste Management Association*. Link [IF = 3]
19. **Henneman, L.R.F.**, Rafaj, P., Annegarn, H., Klausbruckner, C. (2015). "Assessing emissions levels and costs associated with climate and air pollution policies in South Africa." *Energy Policy*. Link [IF = 9]
20. Klausbruckner, C., Annegarn, H., **Henneman, L.R.F.**, and Rafaj, P. (2015). "A policy review of synergies and trade-offs in South African climate change mitigation and air pollution control strategies." *Environmental Science and Policy*. Link [IF = 6]
21. **Henneman, L.R.F.**, Holmes, H., Mulholland, J., Russell, A. (2015). "Meteorological Detrending of Primary and Secondary Pollutant Concentrations: Methods Application and Evaluation Using Long-Term (2000-2012) Detailed Data." *Atmospheric Environment*. Link [IF = 6]
22. Francis, R., Guikema, S., **Henneman, L.R.F.** (2014). "Bayesian Belief Networks for Predicting Drinking Water Distribution System Pipe Breaks." *Reliability Engineering & System Safety*, *130*, 1–11. Link [IF = 7]

Invited Presentations

1. "Air pollution in Virginia: emissions sources and health effects." *Virginia Commonwealth Univesity Symposium of the Institute for Sustainable Energy and the Environment*. April 2026.
2. "Review of the North Omaha Station risk assessment report." Creighton University. December 2025
3. Panelist, *Grad School Life and Application Panel*. Caribbeans for Climate Action. 1 October, 2024.
4. "Mortality from Coal Power Plants in the United States." University of Arizona Department of Chemical and Environmental Engineering Seminar. 23 September, 2024.
5. "Inequitable exposure and mortality from Coal Power Plants in the United States." EPA Office of Air Permitting Brownbag. 7 February, 2024.
6. "Mortality from Coal Power Plants in the United States." Google Geo Sustainability Air Quality Journal Club. 26 June, 2024.
7. "Mortality from Coal Power Plants in the United States." University of Texas Civil and Environmental Engineering Seminar. 20 March, 2024.
8. "Mortality from Coal Power Plants in the United States." GMU-IFEES/GEDC/IUCEE Webinar. 10 April, 2024. Link.
9. "Environmental Health Data Science for More Effective Regulation." *Data Science and AI for public good*. Swiss Federal Statistical Office. 1 December, 2023. Link.
10. "Spatial-temporal trends in four decades of United States on-road air pollution." 10th CRC Mobile Source Air Toxics Workshop. 8-10 February, 2022.
11. "People and power plants: estimating health impacts of energy transitions in the United States." University of Texas, San Antonio Environmental Sciences and Engineering seminar. 10 October, 2020.
12. "People and power plants: estimating health impacts of energy transitions in the United States." Columbia University NIEHS Seminar. 5 December, 2019.
13. "Direct estimates of health improvements from reduced emissions at coal power plants." University of Texas at Austin Environmental Health Interest Group Meeting. Austin, TX. 23 January, 2019.

14. "Direct estimates of health improvements from reduced emissions at coal-fired power plants." EPA Air, Climate, and Energies Centers Work-in-Progress Webinar. 20 September, 2018.
15. "Accountability assessment: The 1990 Clean Air Act Amendments." CRC Mobile Source Air Toxics Meeting. 14-15 February, 2017. Sacramento, CA.

Book Chapters and Reports

1. **Henneman, L.R.F.**, Shen H., Liu, C., Hu, Y., Mulholland, J.A., Russell, A.G. "Ozone in the eastern United States: production efficiency variability over time and between sources." 2020. Air Pollution Modeling and its Application XXVI. https://www.springer.com/gp/book/9783030220549
2. Klausbruckner, C., **Henneman, L.R.F.**, Rafaj, P., Annegarn, H. "Energy policy, regional air quality, and climate mitigation in South Africa: the case for Integrated Assessment." 2018. Applying Systems Analysis to Complex Global Problems. https://link.springer.com/chapter/10.1007%2F978-3-319-71486-8_7
3. **Henneman, L.R.F.**, Liu, C., Chang, H., Lavoué, D., Mulholland, J.A., Russell, A.G. 2016. "Estimating the Impact of Air Pollution Controls on Ambient Concentrations." Air Pollution Modeling and its Application XXIV. http://www.springer.com/us/book/9783319244761
4. Russell, A.G., Tolbert, P., **Henneman, L.R.F.**, Abrams, J., Liu, C., Klein, M., Mulholland, J.A., Sarnat, S., Hu, Y., Chang, H., Odman, T., Strickland, M., Shen, H., Lawal, A. (2017). "Impacts of regulations on air quality and emergency department visits in the Atlanta metropolitan area, 1999-2013." *The Health Effects Institute*. https://www.healtheffects.org/publication/impacts-regulations-air-quality-and-emergency-department-visits-atlanta-metropolitan

Academic Advising

Postdoctoral scholars

- Ting Zhang (2022-2026). Project: "Air pollution source impacts at fine scales for long-term regulatory accountability and environmental justice"

Ph.D. students mentored

- Munshi Md Rasel (2020-2024; graduated Fall 2024)
  Thesis title: "Source specific emissions: Understanding impacts on US air quality and exposure disparities"
- Shreya Guha (2020-Summer 2025; graduated Summer 2025)
  Thesis title: "Effects of meteorology and regulatory policy on ambient pollutant concentrations across China since 2011"
- Xiaorong Shan (2021-present; graduated Summer 2026)
  Thesis title: "Estimating Air Quality from Source-Specific Exposure for Future Application in Environmental Regulations Analyses"
- Sumaiya Hussein (2023-present; passed qualifier Fall 2024)
- Hoda Halaji (2023-present; advanced to candidacy Spring 2026)
- Mohammadreza Bohloul (2024-present; passed qualifier Fall 2025)
- Dinesh Neupane (2024-present; passed qualifier Fall 2025)

M.S. students mentored

- Mohammad Hamza Akhtar (graduated Spring 2026)
  Thesis title: "Linking indoor and outdoor temperature and air quality to building features."

Undergraduate research assistants

- Elizabeth Woosely (2026)
- Sumaia Anwaryan (2024)
- Chikaima Maryjane Udokporo (2024)
- Sara Alhassani (2021)
- Ryah Nadjafi[*‡] (2021-2024)
- Zainab Syed* (2021)
- Gabi Armada (2021)
- Trajan Smeeth (2021)
- Ryan Stauden (2021)

[*]Presented at CEC Undergraduate Research Showcase
[‡]Co-authored a manuscript: **Henneman, L.R.F.**, Ryah Nadjafi, Xiaorong Shan, Jenna Krall. (2025). "Source-specific air pollution emissions inequalities from 2011-2020 in Virginia." *GeoHealth.* [IF = 4.3]

Academic Committees and Other University Service

Mason Fulbright Endorsement Committee (2026-present)

Mason Graduate Interdisciplinary Conference, Discussant (2024, 2026)

CEIE adjunct review committee (2022-2023)

CEIE graduate student engagement coordinator (2022-present)

Mason Bicycle Advisory Committee (2022-present)

PhD Thesis Committee (defense date)

- Felicio Cassalho (2023)
- Ishrat Dollan (2023)
- Azbina Rahman (2021)
- Seth Lawler (2024)

Qualifying Exam

- Ussama Tanveer (2023)
- Janani Kandasamy (2023)
- Kadmiel Adusei (2023)
- Bahman Moghaddame-Jarari (2022)
- Xiaorong Shan (2022)
- Felicio Cassalho (2021)
- Munshi Md Rasel (2021)
- Shreya Guha (2021)
- Meghana Kuppa (2021)
- Allan Rowley (2021)
- Ishrat Dollan (2020)

External Service

- Associate Editor, *Environment International* (2025 – present)
- Associate Editor, *npj Clean Air* (2025 – present)

- Advisory Board Member, Georgia Tech NSF-funded Air Pollution Visualizations for Promoting Data Literacy with Middle Schoolers and the Public (PI: Dr. Jessica Roberts)
- EPA Science Advisory Committee on Chemicals *ad-hoc* member for the for Formaldehyde (March 2024)
- Association for Environmental Engineering and Science Professors (AEESP) Conference Organizing Committee (Conference proposal to host across DC-area institutions, 2025)
- EPA Science Advisory Committee on Chemicals *ad-hoc* member for the for 1,4 Dioxane Supplemental Risk Assessment (September 2023)
- NSF Reviewer Panel, CBET Environmental Engineering Program (March 2023)
- EPA Science Advisory Committee on Chemicals *ad-hoc* member for the TSCA Screening Level Approach for Assessing Ambient Air and Water Exposures to Fenceline Communities Version 1.0 (March 2022)
- EPA Review Panel for Science to Achieve Results (STAR) program [Call: Development of Innovative Approaches to Assess the Toxicity of Chemical Mixtures] (February 2022)
- Reviewer, EPA's TracMyAir mobile app (February 2022)
- Journal Referee (~30 papers/year): *Air Pollution Research; Air Quality, Atmosphere, and Health; American Journal of Epidemiology; Atmosphere; Atmospheric Chemistry and Physics; Arabian Journal of Geosciences; Atmospheric Environment; Energy Policy; Engineering Reports; Environmental Health Perspectives; Environment International; Environmental Impact Assessment; Environmental Policy; Environmental Science and Pollution Research; Environmental Science and Technology; International Journal of Environmental Research and Public Health; International Journal of Epidemiology; The Lancet Public Health; Nature Communications; Nature Energy; One Earth; Open Forum Infectious Disease; JAMA Network Open; Journal of the Air and Waste Management Association; Journal of Environmental Management; Journal of Environmental Policy; Journal of Environmental Studies and Sciences; Journal of Exposure Science and Environmental Epidemiology; Science Advances; Science of the Total Environment*

In the News

1. "New OPPD study says coal is safe in North Omaha but experts raise doubts." *Omaha World Herald*. 14 December, 2025. Link [Quote and analysis provided]
2. "Trump's Misleading Promotion of 'Clean' Coal." *FactCheck.org*. 16 April, 2025. Link [Quote provided]
3. "A solar ban, a gas power plant and the rural retirees firing back at dirty energy." *The Arizona Republic*. 7 April, 2024. Link [Quote provided]
4. Quotes provided for many news outlets regarding paper published in Science, November 2023, including:
    - The New York Times
    - Chicago Tribune
    - Georgia Public Broadcasting
    - Mentioned by John Kerry at COP28 in calls to permit no new coal power plants
5. Contributions to the Virginia Climate Center Blog
    - "Protecting ourselves and our families from wildfire pollution" (June 2023). Written with Jenna Krall, GMU Dept. of Global Community and Health. Link
    - "Proposed power plant in Chesterfield County lands Virginia at a crossroads" (November 2023). Link
6. "Pollution from coal power plants contributes to far more deaths than scientists realized, study shows." *The Conversation*. 23 November, 2023. Link

7. "Air, Lies, and Instagram: Did the air really get better during the lockdown?" *Sierra: The Magazine of the Sierra Club*. 19 May, 2022. Link [Quote provided]
8. "Mason DEVA Project addresses state wide air pollution." *VSE Express*. 15 October, 2022. https://volgenau.gmu.edu/news/2021-10/mason-deva-project-addresses-state-wide-air-pollution
9. "Mason Engineer researcher investigates air pollution's impact on health." *VSE Express*. 1 October, 2020. https://volgenau.gmu.edu/news/588781. [Quote provided]
10. "INSIGHT: On Air Pollution Eastern States Are Right to Point Fingers." *Bloomberg Environment and Energy Report*. 5 December, 2019. https://news.bloombergenvironment.com/environment-and-energy/insight-on-air-pollution-eastern-states-are-right-to-point-fingers. [Op-Ed].

Community

- Board Member, Arlington Families for Safe Streets (2021-present)
- Science Fair Judge, Loudoun County Public Schools Regional Science & Engineering Fair (2021)
- Data science volunteer, Green Streets Initiative (2018)
- Assistant Coach, Chamblee High School Baseball, Chamblee, GA (2014-2017)
- Varsity Baseball, Johns Hopkins University (2008-2012)